MILITARY SPECIFICATION

BOOSTER, M125A1E2 ASSEMBLY LESS BOOSTER PELLET AND LEAD CUP CHARGE

1. SCOPE

1.1 This specification covers one type of booster for use with high explosive ammunition.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-331	- Fuze and Fuze Components Environmental and Performance Tests for
MIL-STD-639	- Surface Roughness Values Preferred for Army Ordnance Materiel
MIL-STD-1167	- Ammunition Data Card

DRAWINGS

Munitions Command

F10520743	- Booster, M125A1E2 Assembly
C7548175	- Box, Packing Ammunition, for Booster Assembly, Details and Packing
C7548174	- Carton Packing
IEL-F10520743	- Inspection Equipment List for Booster, M125A1E2

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:56Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 General.- Materials, parts and assemblies shall comply with the requirements specified on applicable drawings, referenced standards, and specifications except as provided in 6.1.

3.2 Environmental.

3.2.1 The booster shall comply with MIL-STD-331, Jolt, test No. 101.

3.2.2 The booster shall comply with MIL-STD-331, Jumble, test No. 102.

3.2.3 The booster shall comply with MIL-STD-331, Transportation - Vibration, test No. 104.

3.3 Ballistics (See 6.3.2)

3.3.1 The booster shall not function prematurely. (See 6.3.1)

3.3.2 Target impact nonfunction distance.- The booster shall not detonate the round upon impact element function when fired in the 75mm Gun, Service Charge, (2800 feet per second, f.p.s.) on a target 200 plus 5 feet from the muzzle of the weapon.

3.3.3 Target impact function distance.- The booster shall detonate the round upon impact when fired in the 105mm Howitzer, Charge 7 (1550 f.p.s.) on a target 400 minus 5 feet from the muzzle of the weapon.

3.3.4 Ground or water impact.- The booster shall detonate the round upon impact when fired in the 155mm Howitzer, Charge 1 (680 f.p.s.) at an impact angle of 45° minimum.

3.4 Workmanship.- The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.4.1 Metal defects.- All components shall be free from cracks, splits, cold shuts, inclusions, porosity or any similar defect.

3.4.2 Surface finish.- The requirements for surface finish are as detailed on the applicable drawings and MIL-STD-639.

3.4.3 Threads.- Threads shall be full and undamaged for the entire minimum length or depth as required by the applicable drawing.

3.4.4 Burr.- No part shall have a burr which might interfere with the assembly or function of the item of which might be injurious to personnel handling the item.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:56Z
Check the source to verify that this is the current version before use.

3.4.5 Foreign matter.- No part or assembly shall contain chips, dirt, grease, rust, corrosion or other foreign matter. The method and agents used in processing cleaning shall not adversely affect the finish or contaminate the parts.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.2 First article sample.

4.2.1 Initial production sample.- When required by the contract, the first units of production shall be submitted as an initial production sample. The sample shall consist of the following:

40 component parts as delineated in 4.2.2 (aggregate)
100 subassemblies as delineated in 4.2.3 (aggregate)
51 assemblies as delineated in 4.2.5 and 4.2.6 (aggregate)

The sample shall be manufactured using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials, including packaging and packing shall be the same as used for regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and test.- After inspection by the contractor and provisional acceptance at source by the Government, 38 booster assemblies shall be forwarded to a designated proving ground as specified in the contract. Forty (40) component parts, 100 subassemblies and 13 booster assemblies shall be inspected as provided herein, at a Government laboratory or such other facility as specified in the contract.

4.2.2 Components.

4.2.2.1 Twenty (20) of each component shall be tested as in 4.3.3.1 (Hardness).

4.2.3 Subassemblies.

4.2.3.1 Twenty-five (25) of each subassembly shall be tested as in 4.3.3.2 (Static load).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:56Z
Check the source to verify that this is the current version before use.

4.2.4 Acceptance criteria.- Failure of any component or subassembly to comply with the requirements of the applicable Drawing shall be cause for rejection of the initial production sample.

4.2.5 Booster assemblies.- Thirteen (13) booster subassemblies shall be subjected to the following MIL-STD tests:

Test	Quantity	Requirement	Test Method	Acceptance Criteria
Jolt	4	3.2.1	4.4.7.1	4.2.5.1.1
Jumble	4	3.2.2	4.4.7.2	4.2.5.1.1
Transportation-Vibration	5	3.2.3	4.4.7.3	4.2.5.1.2

4.2.5.1 Acceptance criteria.- Occurrence of any of the following in any booster shall be cause for rejection of the initial production sample.

4.2.5.1.1 Jolt and jumble

a. Rotor armed or partially armed. (See 6.2.1 and 6.2.2)

b. Broken, deformed or displaced parts which render the booster unsafe to handle or dangerous to use.

4.2.5.1.2 Transportation-vibration

a. Condition listed in 4.2.5.1.1

b. Failure of the booster to meet the arming and non-arming requirements of the applicable drawing.

4.2.6 Ballistics.- Thirty-eight (38) boosters, of the initial production sample, shall be subjected to the following ballistic test phases.

Phases	Quantity	Requirement	Test Method
Impact Nonfunction Distance	8	3.3.2	4.4.8.1
Impact Function Distance	15	3.3.3	4.4.8.2
Ground or Water Impact	15	3.3.4	4.4.8.3

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:56Z
Check the source to verify that this is the current version before use.

4.2.6.1 Acceptance criteria.- The occurrence of the following shall be cause for rejection of the initial production sample:

a. A premature burst in any phase. (See 6.3.1)

b. Function of one or more rounds upon target impact in the Inpact Nonfunction Phase.

c. Failure of more than two rounds total to function in the combined following phases:

(1) Target Impact Function Phase

(2) Ground or Water Impact Phase.

4.3 Inspection provisions.

4.3.1 Submission of product.- The product shall be presented in accordance with MIL-STD-105.

4.3.2 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with MIL-STD-105. For the classification of defects contained herein, the AQL given shall be used. All nonconforming material shall be rejected. The Ordnance Standard Finish Blocks prescribed in MIL-STD-639 shall be used as a basis of comparison for surface roughness determination. Four power (4X) magnification may be used in performing visual examination for such defects as burrs and foreign matter.

4.3.2.1 Classification of defects.- Inspection of boosters shall be in accordance with the following:

4.3.2.2 Cup, Booster, Drawing B8595509 covering a detail of Drawing F10520743.

Defect		Method of Inspection
CRITICAL		
None defined		
MAJOR	AQL = 1.0%	
101.	Inside depth, improper	Gage
102.	Pitch diameter of thread, improper	Gage
103.	Minor diameter of thread exceeds max.	Gage
104.	Outside diameter, improper	Gage
105.	Effective length of thread, less than min.	Gage
106.	Metal defective	Visual
107.	Cut, score, or gouge on inside wall	Visual
108.	Foreign matter	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:56Z
Check the source to verify that this is the current version before use.

4.3.2.3 Cup, Closing, Drawing B8595508, covering a detail of Drawing F10520743.

Defect	Method of Inspection
CRITICAL	
1. Split, cut or crack	Visual
MAJOR AQL = 1.0%	
101. Length under flange, improper	Gage
102. Outside diameter exceeds max.	Gage
103. Thickness of base	Gage
104. Foreign matter	Visual

4.3.2.4 Body, Drawing F10520746 covering a detail of Drawing F10520743.

CRITICAL	
None defined	
MAJOR AQL = 1.0%	
101. Pitch diameter of large external thread, improper	Gage
102. Major diameter of large external thread, improper	Gage
103. Pitch diameter of small external thread, improper	Gage
104. Major diameter of small external thread, improper	Gage
105. Pitch diameter of internal thread, incorrect	Gage
106. Minor diameter of internal thread exceeds max.	Gage
107. Effective length of internal thread, less than min.	Gage
108. Overall length of body exceeds max.	Gage
109. Diameter of M hole below min.	Gage
110. Staking hole in thread, missing	Visual
111. Burr	Visual
112. Metal defective	Visual

4.3.2.5 Booster, subassembly, Drawing F10520744 covering a detail of Drawing F10520743.

CRITICAL	
1. Interference condition which precludes return of either spring-loaded detent from the fully open position to the position of engagement in the rotor recess. (Check prior to nonarming test)	Visual and Manual
2. Rotor not locked in unarmed position (See 6.2.3) (Check immediately prior to packing)	Visual and Manual

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:56Z
Check the source to verify that this is the current version before use.

4.3.2.5 (Cont'd)

Defect	Method of Inspection
MAJOR AQL = .40%	
101. Excessive protrusion of rotor shaft, pins or screws	Gage
102. Length from base of body to top of plates, exceeds max.	Gage
103. Screw lock pin above flush	Gage
104. Loose cover plate	Visual/Manual
105. Any part missing (excepting detents and springs examined in the above listed critical defect)	Visual
106. Any part broken or deformed (excepting detents and springs examined in the above listed critical defect)	Visual
107. Staking hole missing	Visual
108. Staking missing or inadequate	Visual
109. Burr	Visual
110. Foreign matter	Visual

4.3.2.6 Booster, Assembly F10520743 (less lead charge and booster pellet)

CRITICAL

None defined

MAJOR AQL = 0.40%	
101. Nomenclature missing, incorrect or illegible	Visual

4.3.2.7 Carton, packing, drawing C7548174.

CRITICAL

None defined

MAJOR AQL = 1.0%	
101. Assembly damaged so that contents are exposed or liable to become exposed	Visual
102. Sealing strip incomplete, badly wrinkled, or not in contact with surface of containers for at least 1/4 inch around entire circumference	Visual
MINOR AQL = 2.5%	
201. Contents move when shaken	Manual
202. Marking misleading, incorrect or illegible	Visual
203. Label missing	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:56Z
Check the source to verify that this is the current version before use.

4.3.2.8 Box, packing, drawing C7548175.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR AQL = 1.0%	
101. Board broken	Visual
102. Hardware missing, broken or loose	Visual/Manual
103. Contents exposed	Visual
104. FSC symbol misleading, incorrect or illegible	Visual
105. Strapping missing, broken or loose	Visual/Manual
MINOR AQL = 2.5%	
201. Hardware improperly applied or engaged	Visual/Manual
202. Contents move when shaken	Manual
203. Car seal missing, unsealed or improperly positioned	Visual/Manual
204. Marking misleading, incorrect or illegible	Visual/Manual

4.3.3 Inspection testing.- The following test shall be performed in accordance with the provisions of 4.4.

4.3.3.1 Hardness (Destructive).- Twenty (20) samples from each lot of the following components shall be subjected to the hardness test. Failure of any component to comply with the requirements specified on the applicable drawing shall be cause for rejection of the lot (See 4.4.1).

Component	Drawing Number
Wheel, Escape	C8595539
Pin, Balance	C8595540

4.3.3.2 Static load (Destructive).- Twenty-five (25) samples from each lot of the following subassemblies shall be subjected to the static load test. Failure of any subassembly to comply with the requirement specified on the applicable drawing shall be cause for rejection of the lot of subassemblies, as applicable (See 4.4.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:56Z
Check the source to verify that this is the current version before use.

4.3.3.2 (Cont'd)

Component	Drawing Number
No. 1 Gear and Pinion Assembly	C8594673
No. 2 Gear and Pinion Assembly	C8574681
Escape Wheel Assembly	C8595538
Balance Staff Assembly and Pins	C8594614

4.3.3.3 Non-arming.- Setback Lock Pin (Nondestructive). - Fifty (50) setback lock pin assemblies (see 6.4) assembled to 50 body assemblies from each 4 hours production from each assembly line shall be subjected to the non-arming test. Failure of the setback lock pin to remain above the surface after subjected to the non-arming "G" forces requirement on drawing F10520744, shall be cause for rejection of that 4 hours production of booster assemblies (See 4.4.3).

4.3.3.4 Arming.- Setback Lock pin (Nondestructive).- Fivty (50) setback lock pin assemblies (see 6.4) assembled to 50 body assemblies from each 4 hours production from each assembly line shall be subjected to the arming test. Failure of the setback lock pin to remain locked below the surface after subjected to the arming "G" forces requirement on drawing F10520744, shall be cause for rejection of that 4 hours production of booster assemblies (See 4.4.3).

4.3.3.5 Arming - Booster assembly less setback lock pin assembly (Nondestructive).- Each booster assembly less setback assembly (see 6.4) shall be subjected to the arming test. Failure of the rotor to become locked in alignment with the flash hole, when spun at the Arming Spin Requirement on Drawing F10520744, shall be cause for rejection of the assembly (See 4.4.4).

4.3.3.6 Safety function test (Nondestructive).- Each booster assembly shall be subjected to the safety function test. Failure of both spring loaded detents to open and the rotor to remain locked in the unarmed position by the setback lock pin, when subjected to the Safety Function Spin Requirement on drawing F10520744 shall be cause for rejection of the assembly (See 4.4.5.).

4.3.3.7 Non-Arming (Nondestructive).- Each booster assembly shall be subjected to a non-arming test. Failure of both spring loaded detents to remain locked in the rotor, when subjected to the Non-Arm Spin Requirement on Drawing F10520744, shall be cause for rejection of the assembly (See 4.4.6).

4.3.3.8 Environmental tests (destructive).- Thirteen (13) booster sub-assemblies from each production lot shall be subjected to the following MIL-STD tests (See 4.4.7).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:56Z
Check the source to verify that this is the current version before use.

4.3.3.8 (Cont'd)

Test	Quantity	Requirement	Test Method	Acceptance Criteria
Jolt	4	3.2.1	4.4.7.1	4.3.3.8.1.1
Jumble	4	3.2.2	4.4.7.2	4.3.3.8.1.1
Transportation-Vibration	5	3.2.3	4.4.7.3	4.3.3.8.1.2

4.3.3.8.1 Acceptance criteria.- Occurrence of any of the following in any booster shall be cause for rejection of the production lot.

4.3.3.8.1.1 Jolt and Jumble.

a. Rotor armed or partially armed (See 6.2.1 and 6.2.2)

b. Broken, deformed or displaced parts which render the booster unsafe to handle or dangerous to use

4.3.3.8.1.2 Transportation-Vibration.

a. Conditions listed in 4.3.3.8.1.1.

b. Failure of the booster to meet the arming and nonarming requirements of the applicable drawing.

4.3.3.9 Ballistics (destructive).- Thirty-eight (38) boosters from each lot shall be subjected to the following ballistic test phases: (See 4.4.8)

Phases Ambient	Quantity	Required	Test Method
Impact Nonfunction Distance	8	3.3.2	4.4.8.1
Impact Function Distance	15	3.3.3	4.4.8.2
Ground or Water Impact	15	3.3.4	4.4.8.3

4.3.3.9.1 Acceptance criteria.- The occurrence of the following shall be cause for rejection of the lot the ballistic sample represents:

a. A premature burst in any phase (See 6.3.1)

b. Function of one or more rounds upon target impact in the Impact Nonfunction Phase.

c. Failure of more than two rounds total to function in the combined following phases:

(1) Target Impact Function Phase.

(2) Ground or Water Impact Phase.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:56Z
Check the source to verify that this is the current version before use.

4.3.4 Inspection equipment.- The examinations and tests shall be made using equipment listed on IEL-10520743. Equipment design shall be in accordance with the code designation on the applicable IEL.

4.4 Test method and procedures.

4.4.1 Hardness.- The test shall be performed using the value as specified on the applicable drawing. Samples shall be tested in an area which is solidly supported in order to obtain valid readings. The median value of three hardness readings taken on each sample unit shall be used as the individual reading. All tested samples shall be scrapped.

4.4.2 Static load.- All units shall be tested to the value specified on the applicable drawing. All units subjected to the static load test shall be scrapped.

4.4.3 Arming and non-arming - Setback Lock Pin.- The Setback Lock Pin Assembly (See 6.4) only shall be assembled to the body assembly without staking. All units shall be spun on a centrifuge in accordance with the "G" forces specified on drawing F10520744. After the tests have been performed the setback assembly shall be removed from the body assembly.

4.4.4 Arming - Booster less Setback Lock Pin Assembly.- The unarmed booster assembly prior to assembly of Setback Lock Pin Assembly (See 6.4) shall be spun for a period of 10 seconds maximum in a counterclockwise direction when viewed from the top, at a constant r.p.m. as specified on drawing F10520744. After the test has been performed, the rotor shall be returned to the unarmed position (See 6.2.1 and 6.2.3).

4.4.5 Safety Function Test.- The unarmed booster (See 6.2.3) shall be spun in a counterclockwise direction when viewed from the top, at a constant r.p.m. as specified on drawing F10520744.

4.4.6 Non-Arming.- The unarmed booster (See 6.2.3) shall be spun in a counterclockwise direction when viewed from the top, at a constant r.p.m. as specified on drawing F10520744. The non-arming test shall be last spin test to which the booster is subjected prior to packing.

4.4.7 Environmental Test Procedures.- The component parts of the booster, which have been disassembled and inspected following tests, shall be scrapped.

4.4.7.1 Jolt.- The test shall be performed in accordance with the procedure specified in MIL-STD-331, Test No. 101. The booster shall be disassembled and inspected as follows:

a. Remove dust cover check position of rotor.

b. Check both safety blocks (by probing the blocks for the fully open position to engagement in the rotor recess).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:56Z
Check the source to verify that this is the current version before use.

4.4.7.1 (Cont'd)

c. Remove top plate.

d. Check for broken, deformed or displaced parts (that would render the booster unsafe to handle or dangerous to use).

4.4.7.2 Jumble.- The test shall be performed in accordance with the procedure specified in MIL-STD-331, Test No. 102. The booster shall be disassembled and inspected as follows:

a. Conditions listed in 4.4.7.1

4.4.7.3 Transportation-Vibration.- The test shall be performed in accordance with the procedure specified in MIL-STD-331, Test No. 104.

a. Remove dust cover check position of rotor.

b. Subject the booster to the non-arming and arming requirements in the following sequence:

(1) Spin for booster non-arming requirement

(2) Subject booster to setback lock pin non-arming "G" force requirement, then spin for safety function requirement.

(3) Subject booster to the setback lock pin arming "G" force requirement, then spin for arming requirement.

c. Conditions listed in 4.4.7.1 (c and d)

4.4.8 Ballistics.- The booster assembly, drawing F10520743 shall be assembled to Fuze, PD, M48A3 then assembled to a suitable projectile and fired as follows:

4.4.8.1 Impact nonfunction distance phase.- The test sample shall be fired as specified in 3.3.2 against a suitably supported one inch plywood screen of adequate size under the conditions delineated in 4.2.6 or 4.3.3.9 as appropriate. Observation of impact element function is required.

4.4.8.2 Impact function distance phase.- The test sample shall be fired as specified in 3.3.3 against a suitably supported one inch plywood screen of adequate size under condition delineated in 4.2.6 or 4.3.3.9 as appropriate.

4.4.8.3 Ground or water impact phase.- The test sample shall be fired as specified in 3.3.4 under the conditions as delineated in 4.2.6 or 4.3.3.9 as appropriate.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:56Z
Check the source to verify that this is the current version before use.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, labeling and marking.- Packaging, packing, labeling and marking shall be as specified on Drawing C7548175 with the following exception. Closing Cups, Drawing B8595508 and Booster Cups, Drawing B8595509 may be shipped separately in accordance with best commercial practices as an alternate procedure.

5.2 Data cards.- Data card information shall be as specified in MIL-STD-1167.

6. NOTES

6.1 Components manufactured in strict accordance with applicable drawings represent one set of conditions which will result in satisfactory performance. Dimensions which are specifically required by Classification of Defects, as listed in Section 4 and those which affect external profile of assembly, are mandatory. Pertaining to other dimensions, it is recognized that current mass production methods may occasionally produce parts which are not in full compliance with all dimensional requirements, but which may provide adequate booster performance. Therefore, the contractor, at his own risk, may utilize these parts provided there is no adverse effect on booster functionability. This relaxation of requirements applies to "after the fact" deviations only, to dimensions only, on applicable drawings, and does not constitute authority to establish new dimensions or to relax efforts to achieve the specified limits. The procedure for accomplishing dimensional changes is available under the established engineering change system. Nothing above is to be construed as relieving the contractor or any responsibility in connection with performance of the booster.

6.1.1 The contractor shall maintain records showing deviations from dimensional requirements, the extent of such deviations, the amount of material involved, and the date the material is introduced into the assembly line. Copies of these records shall be forwarded to the agency assigned custody of the technical data.

6.2 Armed/Unarmed Boosters.

6.2.1 Armed.- Rotor is locked by the spring loaded rotor lock pin in the position in which the detonator is fully visible through the flash hole.

6.2.2 Partially armed.- Rotor not locked by both spring-loaded safety blocks.

6.2.3 Unarmed.- Rotor locked by both spring loaded safety blocks with the detonator not at all visible through the flash hole.

6.3 Ballistic Parameters.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:56Z
Check the source to verify that this is the current version before use.

6.3.1 Premature Burst.- Shall be defined as the occurrence of any of the following:

a. If the round detonates in the bore of the weapon

b. If the round detonates in flight within 200 feet from the muzzle of the weapon

6.3.2 Muzzle velocities mentioned in this specification are nominal only and are given for the guidance of the proving ground in selecting appropriate weapons and rounds.

6.4 Setback Lock Pin Assembly.- Shall be defined as the following component parts:

	Drawing No.
Pin, Setback Lock	10534663
Screw, Lock Pin	10534662
Spring, Setback Lock Pin	10534664

6.5 Ordering data.- Procurement documents should specify the title, number and date of this specification.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1390-A606

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:56Z
Check the source to verify that this is the current version before use.